

Probability - Set 2 - Solutions

Specification M7.1-M7.7.

Answer key

Question	Answer	Topic
1	C	Probability

Worked solutions

Question 1 X2-19 · Probability · spec M7.7

Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?

Answer: C

Mutually exclusive gives $P(A \text{ and } B) = 0 \neq P(A)P(B)$, so they cannot be independent: P is sufficient. But two overlapping events can also fail to be independent, so P is not necessary.

Full method - how and why

Relating mutual exclusivity to independence. One implication holds; the converse has easy counterexamples.

- 1 If A, B are mutually exclusive, $P(A \cap B) = 0$. But $P(A)P(B) > 0$ since both probabilities are positive.
why: The positivity hypothesis is what makes the argument work.
- 2 So $P(A \cap B) \neq P(A)P(B)$: they are **not** independent. $P \Rightarrow Q$: **sufficient**.
why: Independence is exactly the equality that fails here.
- 3 Necessity: find dependent events that are **not** mutually exclusive. Roll a die with A : 'even' and B : 'at most 3'.
Then $P(A \cap B) = P(\{2\}) = \frac{1}{6}$, but $P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
why: Unequal, so they are dependent - yet they overlap, so they are not mutually exclusive.
- 4 So P is sufficient but not necessary.
why: Match to the list.

Where it goes wrong

- Treating mutually exclusive and independent as opposites in a way that makes them necessary for each other - dependence has many other causes.
- Forgetting the hypothesis $P(A), P(B) > 0$; with a zero probability, events can be both mutually exclusive and independent.

Takeaway For events of positive probability, mutually exclusive $\Rightarrow$ dependent, but not conversely.